

Development, simulation & implementation of precoding techniques for satellite links

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Master's dissertation submitted in order to obtain the degree of Master of Science in Computer Science Engineering

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Ghent, May 2026

Wannes Baes

Acknowledgements

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by

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Abstract — Laboris magna irure quis sit et veniam voluptate duis elit esse do excepteur. Officia laboris adipisicing laboris duis ut cillum aliquip. Do veniam aute non do cillum sit pariatur consectetur adipisicing qui. Lorem Lorem veniam minim exercitation cupidatat tempor do aute dolore. Nostrud adipisicing tempor quis qui fugiat sint quis ut aute eiusmod Lorem enim veniam. Elit labore Lorem do proident amet anim exercitation culpa enim cillum qui. Aliquip reprehenderit commodo labore minim laborum mollit ut veniam in irure. Minim sit in non aliqua ut elit veniam irure pariatur non velit consequat. Cupidatat veniam occaecat culpa eu irure aute consectetur dolore culpa deserunt quis officia do. Aliqua aliqua eu velit excepteur culpa ea dolore occaecat. Ea est cillum nulla ea cupidatat esse adipisicing ut. Cillum amet esse ex ex enim elit excepteur ullamco in qui aliqua Lorem veniam. Sunt consectetur aliquip commodo exercitation pariatur id tempor adipisicing aliqua. Fugiat cillum ea irure laborum consectetur ipsum. Excepteur laboris eu veniam ex ad consequat esse id ullamco ea cupidatat ad nisi. Aute nisi velit excepteur adipisicing dolore pariatur sunt. Dolore mollit cupidatat esse sint ipsum est nulla nisi eu cupidatat excepteur sint cillum. Culpa irure magna culpa eiusmod ut nisi ipsum veniam elit est occaecat veniam. Sunt adipisicing eu minim qui ea laboris ullamco ut. Aliquip occaecat in occaecat ut tempor dolore sunt consequat qui cillum reprehenderit. Voluptate commodo cupidatat ullamco sint aliquip veniam consequat do. Nisi velit excepteur aute laboris aliqua fugiat minim occaecat quis non incididunt.

Keywords — Precoding Techniques, High-Troughput Satellite Communications, multiple-input multiple-output (MIMO), imperfect channel state information (CSI)

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Fig. 1. Example of a figure caption.

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Symbols and Notations

Abbreviations

acf	autocorrelation function
AWGN	additive white Gaussian noise
BS	base station
BER	bit error rate
CS	circularly symmetric
CSI	channel state information
DFT	discrete Fourier transform
DPC	dirty paper coding
EVD	eigenvalue decomposition
FFT	fast Fourier transform
FIR	finite impulse response
IBR	information bit rate
i.i.d.	independent and identically distributed
KKT	Karush-Kuhn-Tucker
MI	mutual information
MIMO	multiple-input multiple-output
MMSE	minimum mean squared error
MSE	mean squared error
MU	multi-user
pdf	probability density function
PAM	pulse amplitude modulation
PSD	power spectral density

PSK	phase shift keying
QAM	quadrature amplitude modulation
QoS	Quality-of-Service
RX	receiver
SISO	single-input single-output
SNR	signal-to-noise ratio
SR	sum rate
SU	single-user
SVD	singular value decomposition
THP	Tomlinson–Harashima precoding
TX	transmitter
UT	user terminal
WMSE	weighted mean squared error
WMMSE	weighted minimum mean squared error
WSR	weighted sum rate
ZF	zero-forcing

Mathematical Notations

TO DO

System Parameters

N_T	Number of transmit antennas at the BS
N_R	Number of receive antennas at each UT
N_S	Number of data streams per UT
K	Number of UTs

Symbols

TO DO

Part I

Background Material

Chapter 4

Precoding for MU-MIMO systems

Consider the downlink multi-user (MU)-MIMO system introduced in Section 2.2.2. The key difference from the single-user (SU)-MIMO system of the previous chapter is the presence of inter-user interference: the signal intended for one user leaks into the receive antennas of the other users.

The preceding chapter restricted attention to linear precoding, in which the transmitted signal is obtained by applying a linear transformation, represented by a matrix, to the data symbol vector. Nonlinear precoding techniques, on the other hand, apply nonlinear processing to the data symbols before transmission. While linear precoding can achieve the capacity of the SU-MIMO channel, it is no longer capacity-achieving in the MU-MIMO setting.

Two well-known nonlinear precoding techniques are dirty paper coding (DPC) [16] and Tomlinson–Harashima precoding (THP) [17, 18]. DPC is an information-theoretic technique that achieves the capacity region of the Gaussian broadcast channel by encoding each data stream such that interference from all previously encoded streams is pre-subtracted at the transmitter. THP approximates DPC at reduced complexity by successively canceling inter-user interference at the transmitter using modulo arithmetic. Despite their superior capacity, nonlinear techniques incur a significantly higher computational cost. Linear precoding, by contrast, offers near-optimal performance at a fraction of the complexity, and therefore remains the preferred approach in practice. This chapter accordingly focuses on linear precoding techniques.

The following sections presents four commonly used linear precoding strategies for the MU-MIMO downlink, arranged in order of increasing complexity. The presentation begins with three precoding techniques related to ZF, and concludes with the state-of-the-art WMMSE precoder. At the end of this chapter, their performance is compared through numerical simulations for different system configurations. In line with the previous chapter, the design is focused on maximizing the total sum rate (SR) of the system.

4.1 Zero-Forcing Precoding

The objective of zero-forcing (ZF) precoding is to cancel all interference, both inter-user and inter-stream. Moreover, no receiver-side combining is applied. Instead, all processing is done by the precoder and each UT forwards its received samples directly for detection.

Considering these restrictions, we obtain that

$$\mathbf{T} = (\mathbf{G}\mathbf{H}\mathbf{F}) = \sqrt{\mathbf{Q}} \quad \text{and} \quad \mathbf{G} = \mathbf{I}_{\text{rect}}, \quad (4.1)$$

where $\mathbf{T}$ is the compound MU-MIMO transfer matrix, $\sqrt{\mathbf{Q}} \in \mathbb{R}_{>0}^{K N_S \times K N_S}$ is a diagonal scaling matrix and $\mathbf{I}_{\text{rect}}$ is a rectangular diagonal matrix with unit entries on its main diagonal. The compound MU-MIMO input-output relation (2.10) then simplifies to

$$z_{k,i} = q_{k,i} a_{k,i} + n_{k,i} \quad \text{for} \quad k = 1, \dots, K \quad \text{and} \quad i = 1, \dots, N_S. \quad (4.2)$$

This shows that the interference has been completely eliminated and each data stream is received over an independent scalar additive white Gaussian noise (AWGN) channel, which allows performing symbol-by-symbol detection at each UT.

It is shown in [19] that the constraints in (4.1) are satisfied when the precoder is chosen as the right pseudo-inverse of the channel matrix, scaled by $\sqrt{\mathbf{Q}}$:

$$\mathbf{F} = \mathbf{H}^H (\mathbf{H}\mathbf{H}^H)^{-1} \sqrt{\mathbf{Q}} \quad (4.3)$$

This expression requires the channel matrix to have full row rank ($R_H = K N_S$), so that the inverse $(\mathbf{H}\mathbf{H}^H)^{-1}$ exists. For an independent and identically distributed (i.i.d.) Rayleigh fading channel, this condition holds almost surely provided that $N_T \geq K N_S$. In what follows, this condition is assumed to be satisfied. If it is not, the inverse $(\mathbf{H}\mathbf{H}^H)^{-1}$ is replaced by the Moore-Penrose pseudo-inverse, which is computed from the singular value decomposition (SVD) of the channel matrix [20].

It remains to determine the optimal diagonal entries of the scaling matrix $\mathbf{Q}$ so as to maximize the weighted sum rate (WSR) under the total transmit power constraint. Considering (4.2), the WSR of the system can be written as

$$\text{WSR} = \sum_{k=1}^K \sum_{i=1}^{N_S} \mu_{k,i} \log_2 \left(1 + \frac{q_{k,i}}{N_0} \right), \quad (4.4)$$

where $\mu_{k,i}$ is the weight assigned to the i th data stream intended for UT k . Substituting the ZF precoder into the transmit power constraint (2.8) yields

$$\sum_{k=1}^K \sum_{i=1}^{N_S} q_{k,i} ((\mathbf{H}^H \mathbf{H})^{-1})_{n,n} \leq P_t \quad \text{with} \quad n = (k-1)N_S + i. \quad (4.5)$$

Let us introduce the substitution

$$p_n = \gamma_n^{-1} q_{k,i} \quad \text{with} \quad \gamma_n^{-1} = ((\mathbf{H}^H \mathbf{H})^{-1})_{n,n}, \quad (4.6)$$

such that we can write the WSR maximization problem as follows:

$$\begin{aligned} \max_{\{p_n\}} \quad & \sum_{n=1}^{K N_S} \alpha_n \log_2 \left(1 + \frac{\gamma_n}{N_0} p_n \right) \\ \text{s.t.} \quad & \sum_{n=1}^{K N_S} p_n \leq P_t \end{aligned} \quad (4.7)$$

This is exactly the optimization problem handled in Appendix A. The optimal values for p_i are therefore obtained using the waterfilling solution as described in Algorithm 1.

Note that the above power allocation strategy may allocate almost no power to the data streams of a UT with poor channel conditions. The achievable rate for that specific user will therefore be very small, compared the other users with better channel conditions. An alternative, more fair approach is to divide the total transmit power equally across the different users, and use the waterfilling solution to allocate the power to which every user is entitled across the different data streams intended for that user.

To summarize, the steps required for computing the ZF compound precoder matrix for total SR maximization across all UTs are described in Algorithm 1.

Algorithm 1 ZF Precoding

Require:

- The channel matrix: $\mathbf{H}$;
- The total transmit power: P_t .

Ensure: The compound precoder matrix $\mathbf{F}$ and the combiner matrix $\mathbf{G}_k$ for each UT.

- 1: Compute p_i for $i = 1, \dots, K N_S$ using the waterfilling algorithm (Algorithm 1) with $P_t = P_t$, $\alpha_n = 1$, and $\gamma_n = 1/((\mathbf{H}^H \mathbf{H})^{-1})_{n,n}$.
- 2: Compute the entries of the scaling matrix as $q_{k,i} = p_n ((\mathbf{H}^H \mathbf{H})^{-1})_{n,n}$ with $n = (k-1) N_S + i$, for $k = 1, \dots, K$ and $i = 1, \dots, N_S$.
- 3: Compute the compound precoder matrix as $\mathbf{F} = \mathbf{H}^H (\mathbf{H} \mathbf{H}^H)^{-1} \sqrt{\mathbf{Q}}$
- 4: Set the combiner matrices equal the (rectangular) identity matrices $\mathbf{G}_k = \mathbf{I}_{N_S \times N_R}$ for $k = 1, \dots, K$.

Return $\mathbf{F}$, $\mathbf{G}_k$ for $k = 0, \dots, K$

To build an intuitive understanding of ZF precoding, consider a downlink MU-MIMO system with two single-antenna UTs and a BS with two transmit antennas. In this case, the channel matrix is formed by two row vectors, each describing the channel direction toward one user, while the precoder consists of two column vectors, each defining the transmit direction of the data stream intended for a specific UT. The ZF design chooses each precoding vector to be orthogonal to the channel vector of the non-intended user. This ensures that the signal intended for one UT does not leak into the receive antenna of the other UT, thereby eliminating inter-user interference.

This geometry is illustrated in Figure 4.1, where Figure 4.1a corresponds to favorable channel conditions and Figure 4.1b to unfavorable channel conditions.

In the favorable case, the channel directions for the two UTs are nearly orthogonal. As a result, the ZF precoding vectors can remain close to the intended channel directions, which preserves a high effective channel gain. In the unfavorable case, the channel directions are strongly correlated (nearly parallel), so the ZF constraints force the precoding vectors away from the desired channel directions. This substantially reduces the effective gain.

For a fair geometric comparison, the channel vectors in both figures are normalized, and the precoding vectors are scaled such that each stream attains unit effective channel gain. Under this normalization, the much larger precoder norms in the unfavorable case indicate a significantly higher transmit-power requirement to achieve the same effective gain. With a fixed total transmit power in practice, this translates directly into lower received signal-to-noise ratio (SNR) and therefore a lower achievable SR.

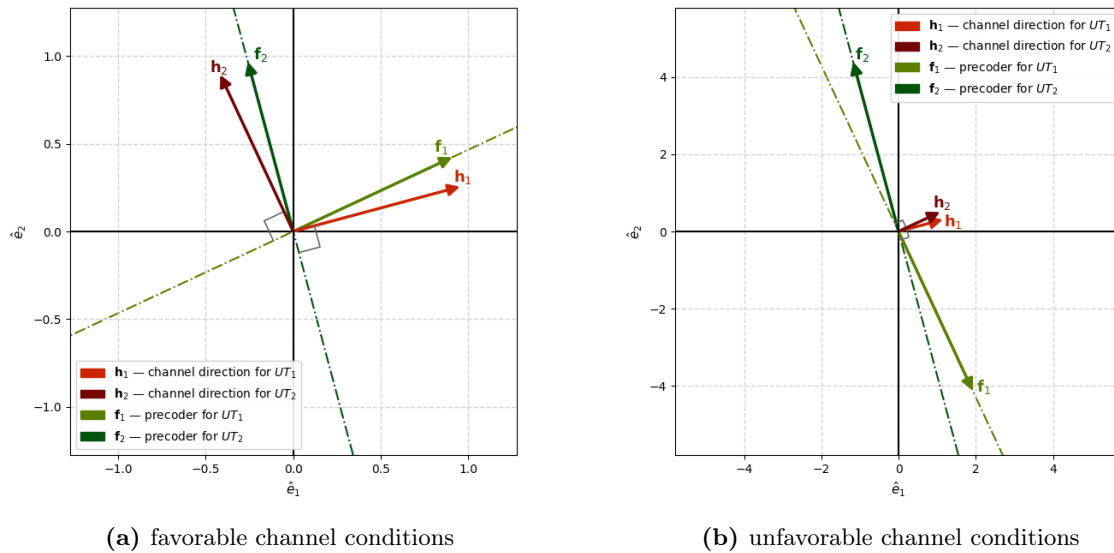


Figure 4.1: Visualization of ZF precoder for a MU-MIMO system with 2 single-antenna UTs and a 2-antenna BS under (a) favorable and (b) unfavorable channel conditions. For both scenarios, the channel vectors are normalized and the precoding vectors are scaled such that the effective channel gain for each data stream is equal to 1. Note the much larger norm of the precoding vectors in the unfavorable scenario compared to the favorable scenario, which illustrates that a lot more power is required to achieve the same effective channel gain when the channel vectors of the UTs are strongly correlated.

Figure 4.2 illustrates what happens when the BS is equipped with an additional transmit antenna while the number of single-antenna UTs remains equal to two. The channel matrix is still formed by two row vectors, one for each user, and the precoder by two column vectors. However, these vectors now lie in a three-dimensional space. As a result, the null-space orthogonal to each user's channel direction is no longer a line but a two-dimensional plane.

In order to eliminate all interference, the precoding vector for each user can be chosen anywhere in the null-space of the other users' channel vectors, that is, in the plane orthogonal to the channel directions of the other users. The best choice is to select the precoding vectors as close as possible to the intended channel directions, which corresponds to projecting the channel vectors onto the

respective orthogonal planes. This maximizes the effective channel gain for each data stream, since it allows as much power as possible to be transmitted in the direction of the intended UT.

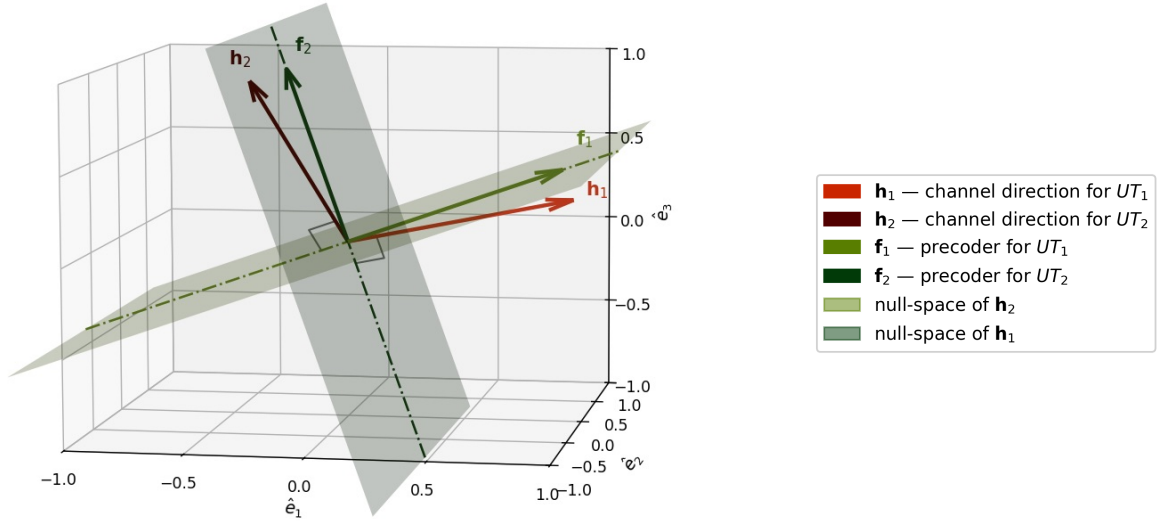


Figure 4.2: Visualization of ZF precoder for a MU-MIMO system with 2 single-antenna UTs and a 3-antenna BS. The channel vectors are normalized and the precoding vectors are scaled such that the effective channel gain for each data stream is equal to 1. With three transmit antennas, the null-space orthogonal to each user's channel direction is two-dimensional. The ZF precoding vector is chosen as the projection of the intended user's channel direction onto this plane, which minimizes the deviation from the desired channel direction while preserving the zero-forcing constraint.

The performance of the ZF precoding technique, in terms of the maximum sum-rate, will be discussed in the next section.

4.2 Zero-Forcing Precoding with Heuristic LSV Combiner

4.3 Block Diagonalization

4.4 Weighted Minimum Mean Square Error

4.5 Performance Comparison

Part II

Precoding for Satellite Communication Systems

Part III

Appendices

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